

Experiment 04 – Conditional Queries

Aim

To understand and implement advanced conditional SQL logic using CASE expressions and complex conditional queries. This lab covers integrating conditional rules within correlated subqueries, CTEs, window functions, and ranking functions, as well as executing targeted data modifications using conditional update statements.

Question 1

(A) (i) You are given a table tbl_happiness that contains the ranking of happiness for different countries. Your task is to retrieve all the records from the table while ensuring that "India" and "Sri Lanka" appear at the top, followed by the rest of the countries in their original ranking order. (B) (ii) Make all the rankings in descending order except India and Sri Lanka.

Answer-A

```
SELECT *
FROM tbl_happiness
ORDER BY
    CASE
        WHEN country = 'India' THEN 1
        WHEN country = 'Sri Lanka' THEN 2
        ELSE 3
    END,
rankings;
```

Output

```
sno | rankings | country
----+-----+-----
  8 |      126 | India
  9 |      128 | Sri Lanka
  1 |         1 | Finland
  2 |         2 | Denmark
  3 |         3 | Iceland
  4 |         4 | Israel
  5 |         5 | Netherlands
  6 |         6 | Sweden
  7 |         7 | Norway
(9 rows)
```

Answer-B

```
SELECT *
FROM tbl_happiness
ORDER BY
    CASE
        WHEN country = 'India' THEN 1
        WHEN country = 'Sri Lanka' THEN 2
        ELSE 3
    END,
rankings DESC;
```

Output

```
sno | rankings | country
-----+-----+-----
8 | 126 | India
9 | 128 | Sri Lanka
7 | 7 | Norway
6 | 6 | Sweden
5 | 5 | Netherlands
4 | 4 | Israel
3 | 3 | Iceland
2 | 2 | Denmark
1 | 1 | Finland
(9 rows)
```

Question 2

A company maintains employee information in a single table where each employee may report to another employee acting as their manager. Prepare a report that displays the details of every employee who supervises at least one other employee. The report should include the manager's employee ID, name, total number of direct reports, and the average age of those reporting employees. The average age should be rounded to the nearest whole number and the final result should be sorted by employee ID.

Answer

```
# SELECT
#     e1.employee_id,
#     e1.name,
#     COUNT(e2.employee_id) AS reports_count,
#     ROUND(AVG(e2.age)) AS average_age
# FROM Employees e1
# LEFT JOIN Employees e2
#     ON e1.employee_id = e2.reports_to
# GROUP BY
#     e1.employee_id,
#     e1.name
# HAVING COUNT(e2.employee_id) > 0
# ORDER BY e1.employee_id;
```

Output

```
employee_id | name | reports_count | average_age
-----+-----+-----+-----
9 | Hercy | 2 | 39
(1 row)
```